

GovMap Native Iframe Tunnel Stability Guide

Problem

The current development route works because CloudFront routes `/govmap-local/*` to a Cloudflare tunnel that points back to the local development backend.

```
https://horizonscanninglab.org/govmap-local/ask
```

This enables GovMap native iframe API calls because the browser origin becomes:

```
https://horizonscanninglab.org
```

That origin is approved by GovMap.

Important Stability Warning

This setup is not stable long-term if it uses a temporary Cloudflare Quick Tunnel hostname like `*.trycloudflare.com`.

The unstable part is not CloudFront. CloudFront is stable. The unstable part is the temporary Cloudflare tunnel hostname.

A Quick Tunnel hostname can change or stop working when:

- `cloudflared` restarts.
- The local machine sleeps or disconnects.
- The tunnel process dies.
- The Quick Tunnel expires.
- A new tunnel is started and receives a different hostname.

When this happens, CloudFront still points to the old tunnel hostname and the public route fails with:

```
502 Failed to contact the origin
```

Short-Term Recovery Procedure

1. Start the local backend.
2. Start a new Cloudflare tunnel to the local backend.
3. Copy the new `*.trycloudflare.com` hostname.
4. Update only the CloudFront origin named `GovMapLocalTunnel`.
5. Keep the cache behavior `govmap-local/*` unchanged.
6. Preserve the CloudFront origin custom headers:

```
X-Forwarded-Proto: https
X-Forwarded-Host: horizonscanninglab.org
```

```
X-Govmap-Proxy-Secret: <existing secret>
```

7. Wait for CloudFront deployment.

8. Verify the public endpoints:

```
https://horizonscanninglab.org/govmap-local/health
https://horizonscanninglab.org/govmap-local/api/govmap/iframe-config
https://horizonscanninglab.org/govmap-local/ask
```

Recommended Stable Solutions

Best Development Solution

Use a named Cloudflare Tunnel instead of a Quick Tunnel. A named tunnel gives a stable hostname and avoids updating CloudFront every time the local tunnel restarts.

Best Production Solution

Deploy the backend/UI directly on AWS behind:

```
https://horizonscanninglab.org
```

Then no local tunnel is needed.

Good Staging Solution

Create a stable staging domain, for example:

```
https://dev.horizonscanninglab.org
```

Then ask GovMap to approve that exact origin.

Important Rule

GovMap native iframe authentication depends on the browser address-bar origin.

For native GovMap iframe API calls to work, the page must be loaded from an approved HTTPS origin such as:

```
https://horizonscanninglab.org
```

It is not enough to proxy only REST calls.